CAPSTONE PROJECT REVIEW PROGRESS RECORD

Review Stage:	**[X] Review–1** [ ] Review–2 [ ] Review–3	**Date of Review:**	**7 August 2026**
Project Title:	Sensitivity-Guided Layer-Wise Approximate Multiplier Mapping for FPGA-Based CNN Accelerators	**Cohort:**	2024–2028 (Batch B4)
Project Guide:	Dr. Arun Nimmala, Associate Professor, Dept. of ECE	**Target HW:**	PYNQ-Z2 (Xilinx Zynq-7020)

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Brief Description of the Project

Briefly explain the premise of the project, the problem being addressed and the proposed approach. (less than 200 words)

Convolutional Neural Network (CNN) edge accelerators are constrained by the high area and power dissipation of Multiply-Accumulate (MAC) datapaths, where 8x8 exact multipliers dominate silicon cost (68 LUTs on Xilinx Zynq-7020). While approximate computing trades controlled arithmetic error for hardware savings, uniform approximation causes severe accuracy degradation or negligible gains. Existing layer-wise mapping methods rely on computationally prohibitive evolutionary search (ALWANN: 7.5 CPU-days) or GPU retraining loops (TRAM). This project proposes a lightweight, single-pass sensitivity metric computed in 0.2 seconds combining static weight variance and dynamic loss gradients via a 256-sample calibration probe without retraining. Layers are ranked such that error-vulnerable early layers receive exact multipliers while noise-resilient deep layers receive aggressive approximate multipliers. The heterogeneous design is mapped onto synthesizable Verilog Processing Elements (PEs) and deployed on a PYNQ-Z2 FPGA board via AXI4-Stream DMA, achieving up to 32.5% multiplier area reduction while preserving baseline classification accuracy.

Progress Achieved up to This Review

Briefly mention the work completed, implementation/simulation/experimental progress, results obtained and other major milestones achieved up to this review.

- **RTL Multiplier Library Design:** Designed four parameterizable 8x8 unsigned multiplier architectures in synthesizable Verilog: exact (full partial-product array), truncated (LSB column drop), compensated (constant midpoint rounding bias), and OR-based logic.
- **Exhaustive 65,536-Pair Simulation:** Verified all 256x256 input combinations in Icarus Verilog; extracted golden arithmetic error distributions (Mean Error, NMED, MRED, Max Absolute Error).
- **Vivado Out-of-Context Synthesis on Zynq-7020:** Obtained post-synthesis FPGA utilization targeting xc7z020clg400-1. Exact: 68 LUTs; Trunc-4: 56 LUTs (-17.6%); Comp-4: 56 LUTs (-17.6%); OR-4: 64 LUTs (-5.9%).
- **Circuit-Level Dominance Discovery:** Proved that Compensated strictly dominates Truncated at identical LUT cost (56 LUTs) with 35% lower error, as the synthesis tool absorbs the +2^(T-1) constant into existing carry logic at zero extra hardware overhead. OR-based design proved inferior and was discarded.
- **Literature Benchmark & Experimental Protocol:** Formulated formal evaluation protocol against ALWANN, TRAM, Sayadi et al., and Kumari & Rao, establishing strict Spearman gate criteria (|rho| >= 0.70) before committing to physical hardware RTL design.

Signature of Team Leader: _____

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